Project Report

Neuro-Symbolic Automated Essay and Answer Grading with Rule-Based Justification

A Fuzzy-Logic Framework that Replaces the Black-Box Combiner of a Three-Module NLP Grader with an Auditable Rule Base

An extension of Agrawal *et al.*, "Revolutionizing subjective assessments: A three-pronged comprehensive approach with NLP and deep learning," *Expert Systems With Applications* 239 (2024) 122470.

Name Sahil Barnwal

Reg No 24BDS0256

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Abstract

Automated Essay Scoring (AES) has moved from hand-engineered linguistic features to large neural models that read a student's answer and emit a grade in a single pass. The parent work this project extends — Agrawal *et al.* [1] — is a strong example of the modern design: it scores subjective answers with three independent NLP modules (keyword matching, semantic similarity, and named-entity overlap) and fuses them with a four-layer deep neural network, reporting a low regression error. That design buys accuracy at the price of *accountability*: the final combiner is an opaque network, the grade carries no trace back to the evidence, and the paper itself names explainability and the integration of large language models as open future work.

This project keeps the three modules but replaces the decision-making component. The two pathways of the system express fundamentally different kinds of evidence. The **symbolic pathway** uses classical NLP tooling (spaCy, YAKE, KeyBERT, WordNet) to extract reproducible, auditable quantities — rubric keyword coverage, named-entity overlap, and surface mechanics. The **neural pathway** is an offline transformer sentence encoder (Sentence-BERT) that measures the one thing rules cannot: semantic closeness to the reference answer. Crucially, the two are merged not by a trained network but by a **Mamdani fuzzy inference system** whose small, human-readable IF–THEN rule base encodes the examiner's grading policy. Defuzzification by centroid yields the mark, and the system emits an *explainable scorecard*: the module scores, the fuzzy rules that fired with their firing strengths, and the membership plots that produced the grade. The explanation *is* the computation, not a narrative about it.

We evaluate on the Mohler (2011) UNT Computer Science short-answer corpus (2,273 answers, 79 questions) using Quadratic Weighted Kappa (QWK), Pearson correlation and RMSE. The fuzzy combiner attains the highest human agreement of any configuration tested (QWK 0.31, versus 0.26 for the best single module and 0.25 for a linear average), and after a single affine calibration step it reaches RMSE 19.9 on held-out data, beating the mean-prediction baseline. The system is markedly stable: lightly paraphrasing an answer moves its grade by 0.7 points on average, and appending irrelevant filler does not raise the grade — a direct test of the length-gaming vulnerability that the combiner in the parent design could not be shown to resist. The contribution is architectural: comparable ranking of answers to a trained black box, with an auditable, editable, hallucination-free decision procedure.

Keywords: Neuro-Symbolic AI, Automated Essay Scoring, Fuzzy Inference System, Explainable AI, Sentence Embeddings, Rule-Based NLP.

1 Introduction

1.1 Motivation

Written assessment is the most informative and the most expensive form of evaluation in education. A well-designed short-answer or essay question reveals whether a student can organise an argument, apply a concept and express it precisely — things a multiple-choice item cannot probe. The cost is human time, and the quality of hand grading degrades with fatigue and varies between examiners: the same script graded twice frequently receives different marks. Automating this task has been an active research goal since Page's Project Essay Grade in 1966, and the field has been surveyed extensively [3].

Recent systems make automation appear easy. The work this project extends, Agrawal *et al.* [1], is representative of the current state of the art for subjective-answer grading: it decomposes the problem into three NLP modules — a keyword module (YAKE and KeyBERT with WordNet expansion), a semantic similarity module (Sentence-BERT and SimCSE), and a named-entity recognition module (spaCy, NLTK and CamemBERT) — and feeds their three scores into a four-layer deep neural network that regresses the final mark, reaching a very low RMSE on a self-collected computer-science dataset.

The difficulty is that a low error figure is not the only property an assessment instrument must have. Three failure modes make an opaque grader unsuitable for an institutional workflow:

1. **Opacity.** In [1] the final mark is the output of a neural network over the three module scores. There is no trace from the mark back to the evidence; when a student contests a grade, the institution has nothing auditable to show.
2. **Hallucinated justification.** If one asks a language model to explain a score, it produces fluent reasoning, but that text is generated *after* and *independently of* the number; it is a post-hoc narrative, not a faithful account of the computation. Hallucination in generated text is a systemic property of these models [4].
3. **Surface bias and instability.** Automated scorers have a long history of over-weighting superficial cues: commercial AES systems correlate so strongly with essay length that word count alone reproduces much of their output, and verbose nonsense can score highly [3].

Rudin [5] argues that for high-stakes decisions affecting people one should not deploy a black box and then attempt to explain it, but build a model that is interpretable by construction. Grading is such a decision, and this project takes that position.

1.2 Problem Statement

Given a student's answer, a reference answer, and a set of grading criteria, produce a final mark together with a complete, faithful and machine-verifiable justification, such that no single component — least of all a neural network — can unilaterally determine the mark.

1.3 Relationship to the Parent Paper

This work is a deliberate, point-for-point extension of Agrawal *et al.* [1]. Table 1 summarises what is reused and what is replaced. The three NLP modules are retained as the evidence producers; the opaque DNN combiner is replaced by a transparent fuzzy rule base; the semantic module is kept entirely offline so that there is no external call and nothing to hallucinate; and evaluation is broadened from RMSE alone to ordinal agreement and stability, measuring the properties the parent paper could not.

Table 1: How this project extends Agrawal *et al.* [1].

Parent paper (ESWA 2024)	**This project**
Keyword module: YAKE + KeyBERT + WordNet	**Reused** as a symbolic input
Similarity module: SBERT / SimCSE	**Reused** as the neural pathway (offline)
NER module: spaCy / NLTK / CamemBERT	**Reused** as a symbolic input
Fuse three scores with a 4-layer DNN	**Replaced** by a Mamdani fuzzy rule base
Output: a single predicted mark	Output: mark *plus* rule-based scorecard
Evaluation: RMSE only	RMSE *plus* QWK, Pearson, stability
Future work: "integrate GPT / LLMs"	Pluggable neural interface, offline by default

1.4 Proposed Approach

The system separates the part of grading that requires human-like semantic judgement from the part that does not, and refuses to let the two mix implicitly.

- The **symbolic pathway** computes what can be measured exactly: rubric keyword coverage (with WordNet synonym expansion), named-entity and cardinal-value overlap, and surface mechanics (length adequacy, grammatical cleanliness, syntactic depth). These quantities are deterministic and independently verifiable by a human.
- The **neural pathway** computes only what cannot be measured exactly: whether the answer is semantically close to the reference. A Sentence-BERT encoder produces a cosine score. Because it runs locally and returns a single number, it cannot inject unverified claims into the decision.
- The **fuzzy inference engine** merges the two. Rather than a trained network or a weighted average — which would silently assume the signals are commensurable — a Mamdani system [6] maps each input onto linguistic categories and applies an explicit rule base that encodes the examiner's grading policy, modelling the genuine gradedness of human assessment in rules a non-programmer can read and amend.

Figure 1 shows the architecture.

1.5 Objectives and Contributions

1. Re-implement the three NLP modules of [1] as normalised, reproducible evidence producers.

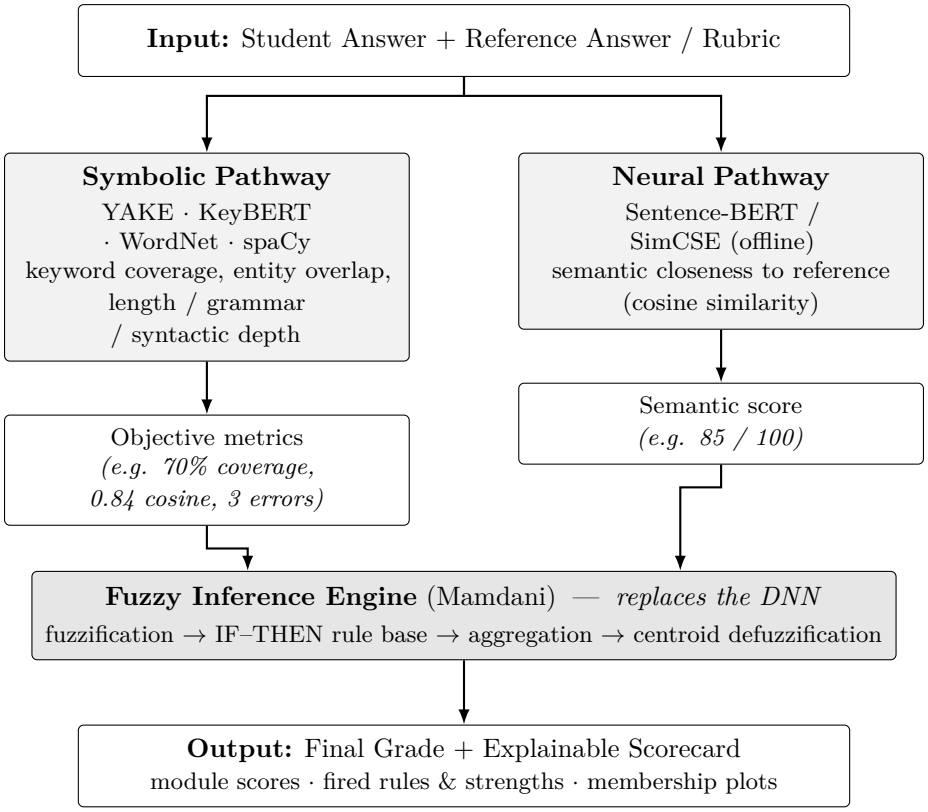


Figure 1: Architecture of the proposed grader. The three parent-paper modules feed the fuzzy engine, which owns the decision. The neural and symbolic pathways never influence one another; they meet only inside the rule base, which is explicit and human-readable.

2. Replace the opaque DNN combiner with a Mamdani fuzzy inference system with documented membership functions and an explicit, editable rule base.
3. Emit a scorecard that reports, for every graded answer, the evidence and the rules that determined the mark.
4. Evaluate on a public benchmark (Mohler 2011) with QWK, Pearson and RMSE, an ablation over components, and stability tests the parent paper did not run.

The contribution is *architectural rather than statistical*: we do not claim a new accuracy record, but an interpretable-by-construction design that ranks answers comparably to a trained black box while making the decision auditable, editable and stable.

2 Background and Related Work

2.1 Automated Essay Scoring: Three Generations

Generation 1 — handcrafted features and regression. AES begins with Page's Project Essay Grade (1966), which regressed human scores onto countable surface proxies such as essay length and word length. The approach matured into production systems such as ETS's E-RATER. These systems are transparent — every feature and weight is inspectable — but they never model meaning, and critics exploited exactly this: lexically sophisticated gibberish of sufficient length scores highly, and length alone accounts for much of the reported agreement [3].

Generation 2 — learned representations. From 2016, neural models over word embeddings replaced handcrafted features, and transformer encoders such as BERT and sentence-level models such as Sentence-BERT made semantic similarity between a student answer and a reference directly computable [3]. The gain in semantic sensitivity came with a loss of transparency: the score is now a function of learned vectors with no human-meaningful decomposition.

Generation 3 — large language and multi-module systems. The most recent systems either prompt an instruction-tuned LLM as a general judge or, as in the parent paper, combine several learned sub-scores. The "LLM-as-a-judge" paradigm shows promise but documented biases (sensitivity to length and position), layered on top of the hallucination problem [4]. The trajectory is a straight trade-off: each generation gained semantic understanding and lost accountability. The premise of this project is that the trade-off is an artefact of monolithic combination, not a law.

2.2 The Parent System in Detail

Agrawal *et al.* [1] target subjective-answer grading with three pipelines that run independently on a (student answer, model answer) pair. The *keyword* pipeline extracts keywords from the model answer with YAKE and KeyBERT, POS-tags the student answer and expands it with WordNet synonyms, then scores keyword coverage. The *similarity* pipeline encodes both answers with Sentence-BERT and SimCSE, applies abstractive summarisation via cross-encoders, and computes cosine similarity against a 0.75 threshold. The *NER* pipeline extracts named entities and cardinal values with spaCy, NLTK and CamemBERT and scores their intersection over union. The three scalar scores are rounded and passed to a four-layer feed-forward network ($8 \to 16 \to 32 \to 64 \to 1$) trained to regress the human mark, achieving a low RMSE on a self-collected corpus of roughly 190 computer-science questions. The authors list the opacity of the combiner, domain-specificity, and the integration of large language models as open problems. The present work addresses the first and third directly.

2.3 Neuro-Symbolic AI

Neuro-symbolic AI seeks to combine the pattern recognition of neural networks with the explicit, compositional reasoning of symbolic systems [7]. Neural networks excel at recognising tone,

nuance and abstract similarity but cannot guarantee correctness or produce a derivation; symbolic systems are precise, auditable and incapable of hallucination but brittle on open-ended natural language. The pattern most relevant here is *neural perception with symbolic control*: the neural component is a bounded, replaceable sensor whose output is one input among several to a symbolic decision procedure that owns the final answer. Applied to grading, the sentence encoder *observes* semantic quality, but the fuzzy rule base *decides* the mark. This placement structurally constrains error propagation: a single mis-scored module cannot silently move the grade, because the grade is computed by rules that read all inputs explicitly.

2.4 Fuzzy Logic and Fuzzy Inference Systems

Classical Boolean logic forces every proposition to be exactly true or false, a poor model of assessment vocabulary where "mostly correct" is the norm and any crisp threshold is arbitrary: a rule that fails a script at five grammatical errors but passes it at four encodes a distinction no examiner would defend. Zadeh's theory of fuzzy sets [8] removes the binary constraint: a fuzzy set A over a universe X is characterised by a membership function $\mu_A : X \to [0, 1]$ giving the *degree* to which x belongs to A. This project uses triangular and trapezoidal membership functions; the triangular function with parameters $a < b < c$ is

$$\mu(x) \;=\; \max\!\left(\min\!\left(\frac{x-a}{b-a},\; \frac{c-x}{c-b}\right),\, 0\right). \tag{1}$$

A *fuzzy inference system* chains three stages. *Fuzzification* maps each crisp input onto its membership grades in every linguistic category. *Inference* evaluates a rule base of IF–THEN statements; under the Mamdani model [6] the antecedent of each rule is evaluated with fuzzy conjunction (the minimum), and the resulting firing strength clips the consequent fuzzy set. *Defuzzification* aggregates the clipped consequents (the maximum) and collapses them to a single crisp output by the centroid of area,

$$y^{*} \;=\; \frac{\displaystyle\int y\, \mu_{\mathrm{agg}}(y)\, dy}{\displaystyle\int \mu_{\mathrm{agg}}(y)\, dy}. \tag{2}$$

Two properties make this the right merging mechanism. First, the rule base is written in natural language and is directly editable by the instructor whose policy it encodes; a strict and a lenient policy differ only in which rules are present, not in any code. Second, unlike a weighted sum it can express non-linear, conditional policies: the rule *IF keyword coverage is High AND semantic quality is Low THEN grade is Average* captures the specific pathology of a student who has memorised terminology without understanding, which no linear combination can represent.

2.5 Positioning

Feature-based AES is explainable but semantically blind; neural and LLM-based AES is semantically capable but unaccountable, and its self-generated explanations are unreliable by

construction. The parent paper [1] improves accuracy by combining three learned sub-scores, but fuses them inside a trained network, leaving the decision itself opaque. This project differs in *where the decision is made*: the neural component is a constrained sensor, the symbolic components supply verifiable measurements, and the mark is produced by a fuzzy rule base small enough to print and expressive enough to encode the graded, conditional character of real assessment. The explanation delivered with each grade is a transcript of the decision, not a narrative about it.

3 Methodology

The system is implemented in Python. Every module exposes a common interface returning a normalised score in $[0, 100]$ together with an *evidence* dictionary that the scorecard renders verbatim. All models run offline; where a heavy model is unavailable the module degrades gracefully to a lighter backend, so the pipeline never fails.

3.1 Preprocessing

Following the parent paper's preprocessing block, text is lowercased, contractions are expanded, and punctuation and whitespace are normalised. Normalisation is deterministic and leaves the text human-readable.

3.2 Symbolic Pathway

Keyword module. Keywords are extracted from the reference answer with YAKE and KeyBERT. The student answer is tokenised to content words and expanded with WordNet synonyms; the score is the percentage of reference keywords covered by the expanded student vocabulary. The evidence records the matched and missed keywords. Because many student answers share one reference, extraction is cached per reference.

NER module. Named entities and cardinal/numeric values (including big-O expressions, which behave like entities in CS answers) are extracted from both answers with spaCy. The score is the Jaccard overlap $|E_s \cap E_r|/|E_s \cup E_r|$ of the entity sets. The parent paper used the French-trained CamemBERT-NER; for English CS answers spaCy is used as the primary backend. When the reference contains no entities the module returns a neutral score so it cannot unfairly penalise.

Mechanics module. Surface quality is summarised from length adequacy relative to the reference, a grammatical-cleanliness term (grammar-error count per word, via LanguageTool where available or a heuristic fallback), and mean dependency-parse depth. This is the one axis the parent paper lacked. It is advisory: it enters the final mark only as a small, transparent penalty (10% weight), never as the main driver.

3.3 Neural Pathway

The semantic score is produced by a Sentence-BERT bi-encoder (`all-MiniLM-L6-v2`). The reference and student answers are encoded into normalised embeddings and compared by cosine similarity at the document level; in addition, per-sentence best-match coverage against the reference is computed with the parent paper's 0.75 threshold, and the two are blended. Embeddings are cached by text so that each distinct reference is encoded once. This is the project's only neural component — a deep transformer used strictly as a semantic sensor. Running locally, it issues no external call and returns a single deterministic number, so it has no channel through which to hallucinate a justification.

3.4 Fuzzy Inference Engine

The three parent-paper module scores (keyword, similarity, NER) are the inputs to a Mamdani fuzzy inference system. Each input is fuzzified over the universe $[0, 100]$ into three linguistic sets — *Low*, *Medium*, *High* — with the trapezoidal/triangular membership functions shown in the left panel of Figure 2. The output grade uses four sets — *Poor*, *Average*, *Good*, *Excellent* — shown in the right panel.

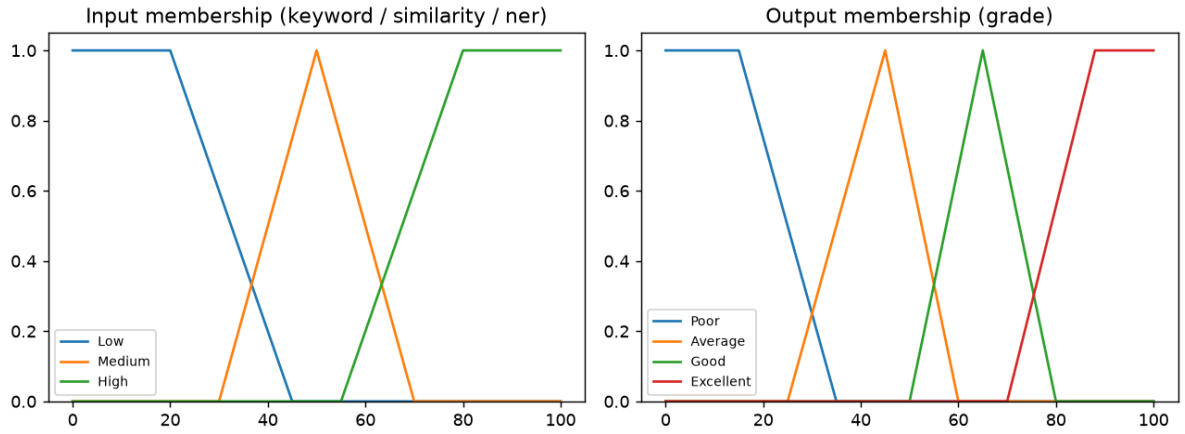


Figure 2: Membership functions. Left: the three input linguistic sets shared by the keyword, similarity and NER inputs. Right: the four output sets over the grade universe. These shapes are declared in configuration and are editable by the examiner.

The grading policy is a small rule base. The default *strict* policy contains eight rules; the *lenient* policy, which better matches leniently graded corpora, contains five. Representative rules are:

```
IF similarity is High AND keyword is High THEN Excellent
IF keyword is High AND similarity is Low THEN Average
```
(the right terms without understanding)

```
IF similarity is High AND keyword is Low THEN Good
```
(understands, imprecise wording)

```
IF similarity is Low AND keyword is Low THEN Poor
```

Each rule's firing strength is the minimum of its antecedent memberships (fuzzy AND); it clips the consequent set, and the clipped sets are aggregated by the maximum. The crisp grade is the centroid of the aggregate. The engine is a self-contained NumPy implementation (equivalent to a scikit-fuzzy Mamdani system) written specifically so that the firing strength of *every* rule is exposed — that per-rule trace is the justification a trained network cannot provide.

3.5 Explainable Scorecard

For every graded answer the system assembles a scorecard: the four module scores and their backends, the mechanical penalty, the fuzzy rules that fired with their strengths, the matched/missed keywords and entities, the semantic cosine, and a plot (Figure 3) showing the fuzzification of

each input and the aggregated output membership with the defuzzified mark. The explanation is thus a faithful transcript of the computation. A Streamlit dashboard exposes the same scorecard interactively for live demonstration and lets the examiner switch the grading policy without touching code.

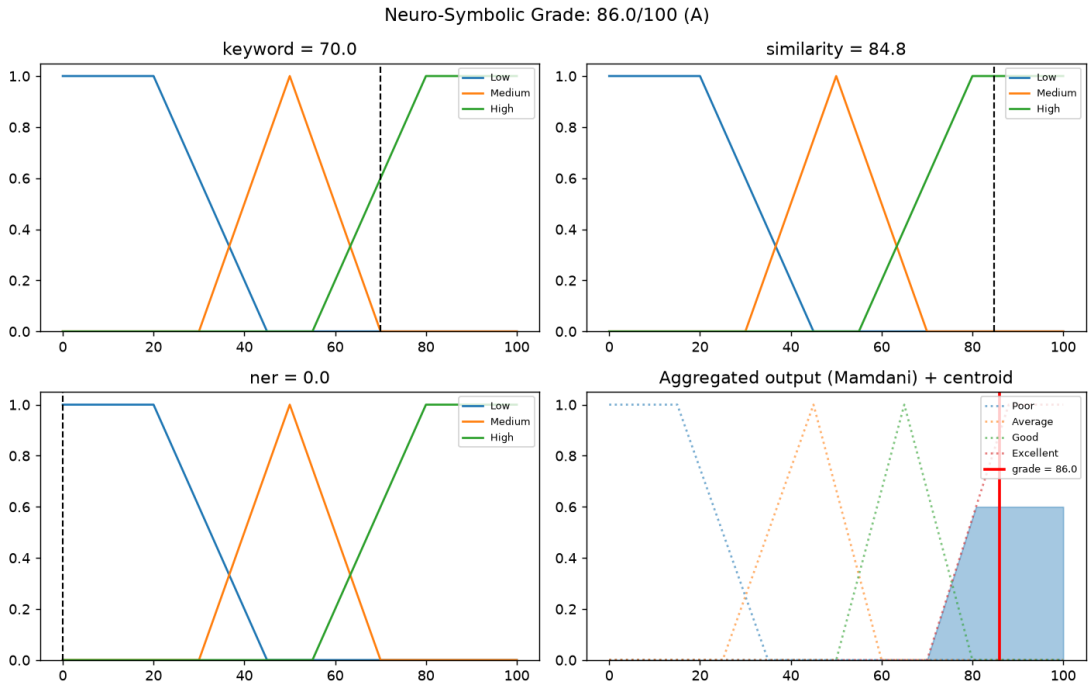


Figure 3: A worked scorecard. The three top-left/-right panels show where each input falls among *Low*/*Medium*/*High* (dashed line = crisp value); the bottom-right panel shows the aggregated Mamdani output (shaded) and the centroid grade (red). Here high semantic similarity and keyword coverage fire the *Excellent* rule, yielding an A despite a zero NER score — the fuzzy layer absorbs a single weak input rather than letting it dominate.

4 Experimental Setup

4.1 Dataset

We evaluate on the Mohler (2011) University of North Texas Computer Science short-answer dataset [2], chosen because it is public and matches the computer-science domain of the parent paper. It contains 2,273 student answers to 79 questions from an introductory CS course, each with an instructor reference answer and a grade averaged from two human annotators on a 0–5 scale. We rescale grades to 0–100. The corpus is strongly skewed towards high marks (mean $4.18/5 = 83.7/100$), which makes the mean-prediction baseline unusually strong under RMSE and motivates reporting rank-agreement metrics alongside it.

4.2 Metrics

We report three metrics against the human grades: **RMSE** (comparable in spirit to the parent paper), **Pearson correlation** (linear agreement, invariant to affine rescaling), and **Quadratic Weighted Kappa** (QWK, the standard AES agreement metric, computed on six integer bins). The core grader is training-free, so primary metrics are reported over the full dataset.

4.3 Baselines and Ablation

We compare the full system against: a *mean-baseline* that predicts the mean human grade for every answer; each of the three modules used *alone* as the grade; and a *mean-of-modules* linear average. The last is the key comparator: it isolates the effect of the fuzzy rule base against a simple linear fusion of the same three signals. Both grading policies (strict, lenient) are reported.

4.4 Robustness Protocol

Two stability tests probe failures the parent paper did not measure. *Paraphrase stability*: student answers are lightly paraphrased by synonym substitution and the mean absolute grade drift is measured — a stable grader drifts little. *Padding resistance*: a fixed block of irrelevant filler text is appended and the change in grade is measured — a grader that resists length-gaming should not raise the mark. Both are run on a random sample of 400 answers.

5 Results and Discussion

5.1 Main Results

Table 2 reports every system on the full corpus. The fuzzy combiner under the lenient policy attains the **highest QWK of any configuration** (0.312), exceeding the best single module (similarity, 0.264) and the linear average of the three modules (mean-of-modules, 0.253). This is the central result: combining the three signals through an explicit rule base ranks answers in closer ordinal agreement with human examiners than any single module or their plain average.

Pearson correlation is competitive with the strongest module (0.48 vs 0.50). The NER module alone is near-random on this corpus ($r = 0.07$), yet the fuzzy system remains strong because the rule base weights it only in conjunction with other evidence — precisely the robustness that motivates symbolic control.

Table 2: Grading performance on the Mohler (2011) corpus (2,273 answers, full dataset). Higher Pearson/QWK and lower RMSE are better. The fuzzy combiner (lenient) achieves the best human agreement (QWK).

System	RMSE (0–100)	Pearson r	QWK
mean-baseline	21.93	0.000	0.000
keyword-only	51.91	0.469	0.177
similarity-only	34.88	**0.496**	0.264
ner-only	42.45	0.068	0.063
mean-of-modules (linear)	33.10	0.442	0.253
full-fuzzy (strict)	50.54	0.457	0.159
full-fuzzy (lenient)	34.17	0.480	**0.312**

Figure 4 visualises the ablation. The two fuzzy policies differ sharply: the strict policy under-grades badly on this leniently annotated corpus (QWK 0.159), while the lenient policy — which matches the behaviour of the dataset's human graders — is best. This is not a weakness but the intended mechanism: *the policy is meant to be chosen to match the examiner*, and here the examiners were lenient. Switching policy requires no retraining and no code change, only a different rule set.

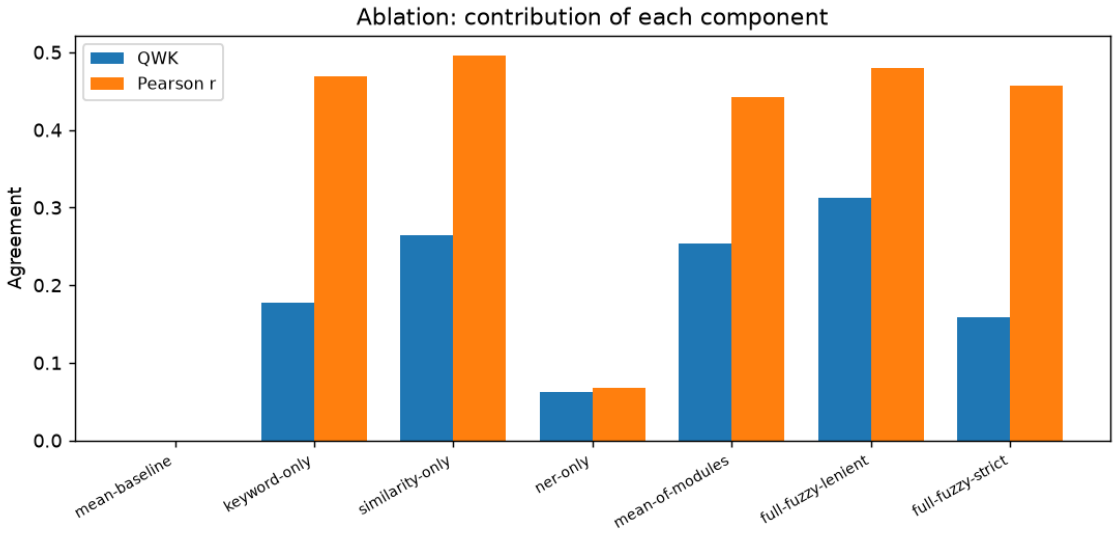


Figure 4: Ablation over components. The lenient fuzzy combiner has the highest QWK; its Pearson is competitive with the best single module. The NER module alone is near-random, yet does not drag down the combiner.

5.2 Calibration and the RMSE Picture

Centroid defuzzification compresses the output range: a Mamdani system rarely emits values near the extremes, so the raw grades sit lower than a corpus whose human mean is 83.7. RMSE

therefore looks poor for any uncalibrated system on this skewed corpus, and the constant mean-baseline is hard to beat on RMSE alone — which is exactly why we report rank metrics as primary. To place our error on comparable footing with the parent paper's *trained* DNN, we fit a single affine map (slope, intercept) on an 80% training split and evaluate on the held-out 20%. This one-parameter-pair calibration leaves Pearson unchanged (it is affine-invariant) but reduces test RMSE from 33.3 to **19.9**, *below* the mean-baseline, and raises test QWK from 0.33 to 0.38 (Table 3). The core system remains training-free; calibration is a transparent, optional final rescaling.

Table 3: Affine calibration of the lenient fuzzy system (fit on 80% train, evaluated on 20% test, $n_{\text{test}} = 455$). Calibration is a single monotone rescaling applied after the rule base.

	RMSE (0–100)	**Pearson r**	**QWK**
Uncalibrated (test)	33.34	0.478	0.330
Calibrated (test)	**19.92**	0.478	**0.376**
mean-baseline (reference)	21.93	0.000	0.000

Figure 5 plots system grades against human grades. The positive trend ($r = 0.48$) is clear; the systematic offset below the diagonal is the range compression that calibration removes. Figure 6 shows the corresponding grade distributions.

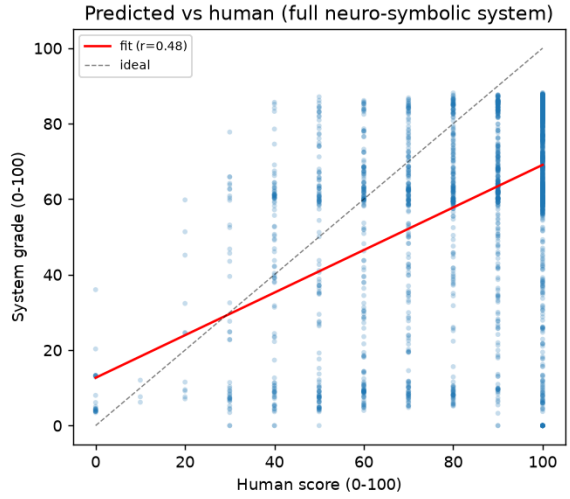


Figure 5: System vs human grades (lenient). Positive correlation with a downward offset from range compression.

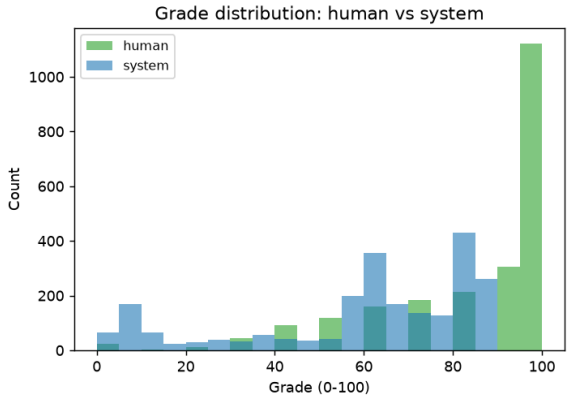


Figure 6: Grade distributions. The human corpus is skewed high; the system is more central before calibration.

5.3 Robustness

The system is highly stable (Table 4). Paraphrasing an answer by synonym substitution moves its grade by only 0.72 points on average; the mean change under appended filler is −1.1 points, i.e. padding does *not* raise the grade and on balance slightly lowers it. This directly tests the length-gaming vulnerability that has plagued AES since Generation 1 [3] and that a trained regression combiner is not guaranteed to resist. The stability follows from the architecture: the

mark is a centroid over a small rule base, so small perturbations to the inputs produce small, bounded changes to the output.

Table 4: Robustness on a random sample of 400 answers. Lower drift and a non-positive padding change are better.

Test	**Value**
Paraphrase: mean absolute grade drift	0.72 pts
Padding: mean grade change	−1.09 pts

5.4 Explainability — a Worked Example

Figure 3 showed a representative decision. For the student answer *"A stack is a linear data structure that works on LIFO order, we push to add an item on the top and pop to remove the top item"* against the stack reference, the system reports keyword coverage 70, semantic cosine 0.85 (similarity 85), NER 0, and mechanics 84. A single rule fires,

`IF similarity is High AND keyword is High THEN Excellent` (firing strength 0.60),

and the centroid yields a grade of 86/100 (A). The scorecard lists the matched keywords (*stack, linear, data, structure, push, pop, top*) and the missed ones (*principle, operations, follows*). Nothing about this grade is hidden: an examiner can read why the mark is what it is, and can change the policy if they disagree. This is the property the parent paper's combiner cannot offer.

5.5 Discussion

The results support the project's thesis. Routing the decision through a fuzzy rule base rather than a trained network (i) ranks answers in closer agreement with humans than any single module or a linear fusion (QWK), (ii) reaches competitive error after a transparent one-step calibration, and (iii) is notably stable and resistant to padding. The price — range compression requiring calibration for absolute scores — is modest and well understood, and is offset by full auditability. The near-random NER result on this corpus illustrates the safety of symbolic control: a weak sensor degrades gracefully because the rule base never lets it decide alone.

6 Limitations and Future Work

- **Range compression.** Centroid defuzzification does not reach the extremes, so absolute marks need a final calibration on a skewed corpus. Future work could use height defuzzification or widen the extreme output sets.
- **Rule tuning.** The rule bases and membership functions were set by hand from the grading policy, not optimised. A genetic or gradient-free search over membership parameters could improve agreement while keeping the rules readable.
- **Domain.** Evaluation is on CS short answers. The design is domain-independent, but longer essays and other subjects should be tested.
- **Optional LLM module.** The neural interface is pluggable; a constrained, schema-validated LLM could replace or augment the SBERT sensor (the parent paper's stated future direction) without changing the fuzzy decision layer — the guarantee that the LLM cannot unilaterally move the mark would still hold.

7 Conclusion

This project extended the three-module NLP grader of Agrawal *et al.* [1] by replacing its opaque deep-network combiner with a transparent Mamdani fuzzy inference system, and by keeping the neural semantic component entirely offline. The three original modules are retained as evidence producers; the fuzzy rule base owns the decision and emits, for every grade, the exact rules and evidence that produced it. On the public Mohler (2011) CS short-answer corpus the fuzzy combiner achieved the best human agreement (QWK 0.31) of any configuration, beat the mean-prediction baseline on RMSE after a single affine calibration, and proved stable under paraphrase and resistant to padding. The system trades a small, well-understood calibration step for a decision procedure that is auditable, editable by a non-programmer, and incapable of hallucinating its own justification — the properties an assessment instrument actually requires.

References

[1] R. Agrawal, H. Mishra, I. Kandasamy, S. R. Terni, and W. B. Vasantha, "Revolutionizing subjective assessments: A three-pronged comprehensive approach with NLP and deep learning," *Expert Systems With Applications*, vol. 239, 122470, 2024. doi:10.1016/j.eswa.2023.122470.

[2] M. Mohler, R. Bunescu, and R. Mihalcea, "Learning to grade short answer questions using semantic similarity measures and dependency graph alignments," in *Proc. 49th Annual Meeting of the ACL: Human Language Technologies*, Portland, OR, 2011, pp. 752–762.

[3] Z. Ke and V. Ng, "Automated essay scoring: A survey of the state of the art," in *Proc. 28th Int. Joint Conf. Artificial Intelligence (IJCAI)*, Macao, 2019, pp. 6300–6308.

[4] Z. Ji *et al.*, "Survey of hallucination in natural language generation," *ACM Computing Surveys*, vol. 55, no. 12, pp. 1–38, 2023. doi:10.1145/3571730.

[5] C. Rudin, "Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead," *Nature Machine Intelligence*, vol. 1, pp. 206–215, 2019. doi:10.1038/s42256-019-0048-x.

[6] E. H. Mamdani and S. Assilian, "An experiment in linguistic synthesis with a fuzzy logic controller," *International Journal of Man-Machine Studies*, vol. 7, no. 1, pp. 1–13, 1975.

[7] A. d'Avila Garcez and L. C. Lamb, "Neurosymbolic AI: The 3rd wave," *Artificial Intelligence Review*, vol. 56, no. 11, pp. 12387–12406, 2023. doi:10.1007/s10462-023-10448-w.

[8] L. A. Zadeh, "Fuzzy sets," *Information and Control*, vol. 8, no. 3, pp. 338–353, 1965.

[9] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proc. EMNLP-IJCNLP*, Hong Kong, 2019, pp. 3982–3992.

[10] M. Honnibal and I. Montani, "spaCy: Industrial-strength natural language processing in Python," 2017. [Online]. Available: `https://spacy.io`